H3-PSNET: Hierarchical Point SAM with Contrastive Refinement and Risk-Aware Distillation for 3D Part Segmentation

A report submitted in partial fulfilment of the requirements for the award of the degree of B.TechComputer Science and Engineering

by

Ananya Srivastava
(Roll No: 122CS0077)

& **Harshith TNV**
(Roll No: 122CS0040)

Under the Guidance of

Dr. Shounak Chakraborty
Assistant Professor
Dept. of Computer Science and Engineering

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
INDIAN INSTITUTE OF INFORMATION TECHNOLOGY DESIGN AND MANUFACTURING KURNOOL

Evaluation Sheet

Title of the Project: H3-PSNET: Hierarchical Point SAM with Contrastive Refinement and Risk-Aware Distillation for 3D Part Segmentation

Name of the Student: Ananya Srivastava

(122CS0077)

Harshith TNV

(122CS0040)

Examiner(s):

T C. Nee

Supervisor/Advisor:

Shounak
24/Apr/2026

Head of the Department:

Srinivas Naik

Date:

Place:

Declaration

We, **Ananya Srivastava** and **Harshith TNV**, with Roll No: **122CS0077** and **122CS0040** respectively hereby declare that the material presented in the Project Report titled **H3-PSNET: Hierarchical Point SAM with Contrastive Refinement and Risk-Aware Distillation for 3D Part Segmentation** represents original work carried out by us in the **Department of Computer Science and Engineering** at the **Indian Institute of Information Technology Design and Manufacturing Kurnool** during the year **2025-2026**.

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Date: 24/4/26

Ananya

Ananya Srivastava (122CS0077)

Student's Signature

Harshith TNV (122CS0040)

Harshith

Student's Signature

In my capacity as supervisor of the above-mentioned work, I certify that the work presented in this Report is carried out under my supervision, and is worthy of consideration for the requirements of B.Tech. Project work.

Advisor's Name: **Dr. Shounak Chakraborty**

Shounak 24/April/2026

Advisor's Signature

Assistant Professor
Department of Computer Science & Engineering
IIITDM Kurnool
Jagannathagattu, Dinnedevarapadu (V)
Kurnool-518008, Andhra Pradesh, INDIA.

Abstract

The adaptation of foundational models like Meta AI's Segment Anything Model (SAM) to 3D point clouds presents significant challenges in resolving geometric ambiguity and boundary confusion. Unlike 2D natural images or Remote Sensing Imagery (RSI), which rely on dense pixel grids and spectral signatures, 3D point clouds are characterized by non-uniform density, lack of inherent order, and irregular geometric structures. This fundamental domain gap renders existing methods such as PointSAM and SAMNet++ insufficient, as the transition from 2D priors to 3D refinement frequently results in substantial contextual information loss and ineffective class-agnostic prompting. Furthermore, geometry-based approaches like 3D-AOCL (3D Analytical Online Continual Learning) suffer from class-agnostic sampling and density neglect, while standard neural architectures like PointNet exhibit feature space dominance and biased decision boundaries. In high-precision applications like autonomous navigation or robotics, the misidentification of minority objects is more than a nuisance; it leads to catastrophic failures in scene understanding and safety.

To address these inefficiencies, we propose H3-PSNet (3-Headed PointSAMNet), a novel hierarchical framework for imbalance-aware 3D semantic segmentation. Our model introduces three specialized heads: (1) Negative Point Contrast (NPC), which utilizes subtractive prompting to push decision boundaries away from spatial confusion zones; (2) Prototype-Based Regularization (PBR), which employs Hungarian Matching to align batch features with global semantic prototypes ; and (3) a Geometric Risk Head powered by Inverted μ-Relief logic. By inverting the Relief algorithm, we effectively distill geometric complexity into a learning signal, prioritizing "hard" boundary points where hits and misses are nearly indistinguishable.

We evaluate H3-PSNet on ModelNet40, ShapeNet Part, and S3DIS Area5, where it demonstrates superior convergence and robust performance on minority categories. Beyond standard benchmarks, H3-PSNet offers significant potential for real-time LiDAR streams and uncertainty-guided semi-supervised learning.

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Date: Ananya Srivastava

HarshithTNV

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List of Abbreviations

3D-AOCL	3D Analytical Online Continual Learning
ADASYN	Adaptive Synthetic Sampling
BCE	Binary Cross-Entropy
CAD	Computer-Aided Design
CNN	Convolutional Neural Network
DDPM	Denoising Diffusion Probabilistic Model
DGCNN	Dynamic Graph Convolutional Neural Network
EMA	Exponential Moving Average
FP	Feature Propagation
FPS	Farthest Point Sampling
IoU	Intersection over Union
LiDAR	Light Detection and Ranging
LoRA	Low-Rank Adaptation
mAcc	Mean Class Accuracy
mIoU	Mean Intersection over Union
MLP	Multi-Layer Perceptron
MSE	Mean Squared Error
NLP	Natural Language Processing
NPC	Negative Point Contrast
oA	Overall Accuracy
OHEM	Online Hard Example Mining
OOD	Out-of-Distribution
PBR	Prototype-Based Regularization
ROS	Random Oversampling
RSI	Remote Sensing Imagery
SA	Set Abstraction
SAM	Segment Anything Model
SMOTE	Synthetic Minority Oversampling Technique

Chapter 1

Introduction

1.1 Overview

In high-precision applications such as autonomous navigation and robotics, the accurate identification and parsing of 3D environments is critical. Misidentifying objects or boundaries in these domains is more than a minor inconvenience; it can lead to catastrophic failures in scene understanding and overall safety. As a result, robust 3D scene parsing remains a primary focus in modern computer vision research.

1.2 Semantic Segmentation in Point Clouds

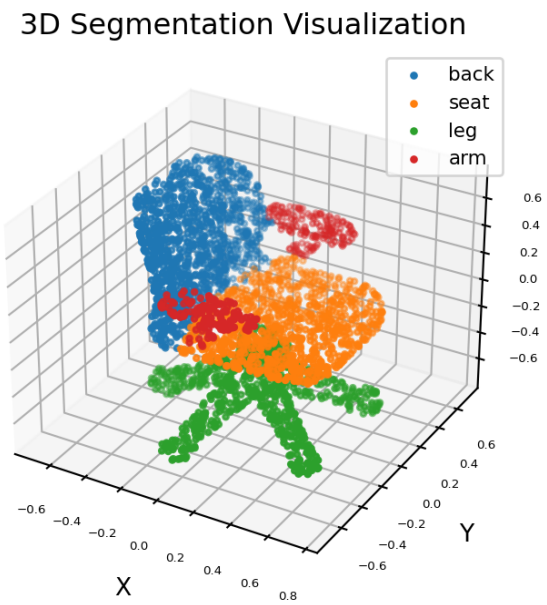

Figure 1.1: Segmenting a 3D scene using PointNet where the model distinguishes between "Chair legs," "Seats," 'Arms," and "Back"

Semantic segmentation in 3D point clouds is inherently more complex than its 2D counterpart. Unlike 2D natural images or Remote Sensing Imagery (RSI), which rely on dense, highly structured pixel grids and spectral signatures[1], 3D point clouds possess irregular geometric structures[3][4]. They are characterized by a lack of inherent order

and non-uniform density. These properties render native processing difficult and make it challenging for networks to accurately capture fine-grained local structures and geometric boundaries.

1.3 Challenges in Semantic Segmentation of Point Clouds

A constant challenge in 3D semantic segmentation is severe class imbalance, where minority classes (such as motorbikes in street scenes or small structural elements in indoor spaces) are vastly underrepresented compared to majority background elements. Standard deep learning strategies often fall victim to "inertial learning," a phenomenon where the large volume of majority-class examples dominates the feature space and gradient direction during early training. Consequently, the network settles into a local minimum, neglecting the subtle geometric features of rare classes. Traditional data-level fixes like Random Oversampling (ROS) [14]or SMOTE[16] often fail in 3D, as duplicating points destroys local density cues, and interpolation creates geometric noise that corrupts the surface manifold.

The Segment Anything Model (SAM), developed by Meta AI, has emerged as a powerful foundation model that shifts computer vision away from task-specific architectures toward promptable interfaces. Trained on a massive dataset of masks and prompts, SAM exhibits exceptional zero-shot segmentation capabilities on 2D images[25], guided by sparse positive prompts like points or bounding boxes. Its primary strength lies in resolving visual confusion through interactive prompt refinement, which has led to widespread adoption in 2D fields like medical imaging and remote sensing.

1.3.1 Foundational Models in Semantic Segmentation

Historically, the field of computer vision has relied heavily on task-specific architectures, where deep learning models were trained from scratch or fine-tuned on closed-set datasets for highly specific applications. While these conventional models achieved commendable performance within their designated domains, they lacked the flexibility to generalize to unseen classes, irregular geometries, or novel environments without requiring exhaustive retraining and domain-specific annotations.

This paradigm has been fundamentally disrupted by the advent of foundation models. Foundation models are versatile, large-scale neural networks trained on massive, diverse datasets, designed to serve as a robust, generalized base for a wide array of downstream tasks. Following the unprecedented success of foundation models in Natural Language Processing (NLP), such as BERT and GPT-3, as well as in multimodal zero-shot tasks like CLIP, the computer vision community has rapidly shifted its focus toward generalized, promptable architectures.

In the realm of semantic segmentation, the Segment Anything Model (SAM), developed by Meta AI, represents a major architectural breakthrough. Trained on a billion-scale dataset of images and high-quality masks, SAM introduced a highly adaptable, promptable interface to pixel-level classification. Instead of passively processing an image to find pre-defined classes, SAM interacts with sparse positive and negative prompts, such as spatial points, bounding boxes, or text to dynamically generate and refine segmentation masks. Its exceptional zero-shot capabilities allow it to resolve visual ambiguity through iterative prompt refinement, leading to its widespread adoption across various 2D

domains, including medical imaging, autonomous driving[23][24][26][27][28], and remote sensing[29].

However, despite the immense success of foundational models like SAM in 2D image processing, their direct application to the 3D domain remains a complex and largely unresolved challenge. While 2D foundation models leverage dense, structured pixel grids and rich spectral information, 3D point clouds are inherently unstructured, sparse, and lack a fixed topological order. Bridging this domain gap, adapting the robust zero-shot priors and promptable nature of 2D foundation models to natively understand the geometric irregularities and spatial ambiguities of 3D point clouds is a central focus of modern segmentation research and serves as a primary motivation for the proposed hierarchical framework.

1.3.2 Spatial Ambiguity and Subtractive Prompting

In the context of 3D point cloud processing, spatial ambiguity refers to the inherent uncertainty encountered when determining the precise boundaries between dense, geometrically similar, or physically adjacent object parts. Unlike 2D imagery, where explicit color contrasts and texture gradients often delineate distinct objects[25], 3D parts frequently share smooth, continuous surface manifolds. Consequently, the semantic separation of these parts relies entirely on subtle shifts in curvature or point density, making the exact point of transition highly ambiguous.

To counteract this phenomenon, the concept of subtractive prompting is utilized. Subtractive prompting is a targeted intervention strategy designed to establish negative constraints within a model's latent feature space. Rather than guiding the network solely by indicating where a target object is located, this mechanism explicitly dictates where the object is *not*. By injecting these negative spatial constraints, it generates a repulsive field around the target's perimeter, actively forcing the decision boundary away from interfering zones.

Understanding these definitions highlights a critical hurdle in current point-supervised 3D segmentation frameworks. When navigating spatial ambiguity, providing a single positive prompt on a target part frequently causes the model to "bleed" the predicted mask into neighboring structures due to their sheer spatial proximity. Standard foundation model adaptations generally lack the native geometric understanding necessary to separate these interconnected parts based solely on localized curvature or surface normals. Relying exclusively on positive prompts fails to define these complex boundaries effectively, necessitating the subtractive prompting mechanism to explicitly limit the network's receptive field and resolve the semantic confusion.

1.3.3 Geometric Uncertainty and Hard Example Mining

The concept of "Hard Example Mining" is well-established in 2D computer vision, frequently utilized in object detection frameworks via Online Hard Example Mining (OHEM) [11]to force models to learn from the examples they struggle with the most. However, directly translating this concept to unstructured 3D space introduces several highly complex challenges:

- **Ambiguity of "Difficulty":** In 3D point clouds, defining what makes a point "difficult" is intrinsically complicated. A point may be hard to classify due to se-

mantic rarity (belonging to an underrepresented minority class) or due to geometric difficulty (existing in a noisy, sparse area)[12][13].

- **Vulnerability to Sensor Noise:** Standard OHEM methods often conflate these two types of difficulty[15]. This conflation causes training instability, as the network frequently ends up over-optimizing on spatial outliers or sensor artifacts rather than learning true structural boundaries.

- **The Boundary vs. Center Discrepancy:** Standard training approaches treat all points equally. This ignores the geometric reality that boundary points are inherently harder to classify than the central points of an object.

- **Computational Overhead of Uncertainty Estimation:** Traditional methods for quantifying uncertainty, such as Bayesian approximations or Monte Carlo Dropout, introduce significant inference latency, making them impractical for dynamic training on massive 3D datasets.

Properly distilling this geometric uncertainty into a distinct, efficient learning signal is required to force networks to prioritize geometrically confusing regions where decision boundaries are blurred, without falling victim to outlier overfitting.

1.4 Problem Statement

Current foundation model adaptations for 3D point clouds struggle to resolve geometric ambiguity and boundary confusion, particularly when utilizing cost-effective point-level supervision. The transition from 2D priors to 3D spaces often results in contextual information loss and ineffective class-agnostic prompting, leading to spatial mask bleeding. Furthermore, existing 3D networks fail to adequately address severe class imbalance, succumbing to feature space dominance by majority classes and neglecting geometrically confusing boundary points of minority objects

1.5 Contribution of the Thesis

The primary objectives of this research are as follows:

- To adapt 2D foundation model priors (SAM) to complex 3D point cloud segmentation using highly efficient, point-level supervision.

- To design a subtractive prompting mechanism that dynamically resolves spatial conflicts and mask bleeding in dense, adjacent 3D parts.

- To overcome class imbalance by transforming local geometric confusion into a targeted supervision signal, thereby forcing the network to focus on hard-to-classify boundary points.

- To maintain global semantic consistency across imbalanced batches, preventing feature drift during localized boundary refinement.

1.6 Organisation of the Thesis

The remainder of this thesis is organized as follows: Chapter 2 provides a comprehensive review of related literature, encompassing foundation models, 3D adaptation techniques, and imbalance learning strategies. Chapter 3 details the proposed methodology, outlining the architecture of H3-PSNet and its three specialized heads. Chapter 4 presents the experimental setup, dataset descriptions, and quantitative/qualitative evaluations against state-of-the-art baselines. Finally, Chapter 5 concludes the report and discusses potential avenues for future research.

Chapter 2

Literature Survey

2.1 Foundation Models and 3D Adaptation

The paradigm of computer vision has recently shifted from task-specific architectures to generalized, promptable foundation models. The Segment Anything Model (SAM), developed by Meta AI, represents a significant leap in this domain. Trained on a billion-scale dataset of images and masks, SAM demonstrates exceptional zero-shot segmentation capabilities across various downstream tasks, driven by sparse prompts such as points or bounding boxes. However, transitioning these 2D-centric foundation models to the 3D domain introduces profound challenges. Unlike 2D images, which rely on dense, highly structured pixel grids, 3D point clouds are characterized by irregularity, sparsity, and a lack of inherent topological order. Early attempts to bridge this domain gap primarily relied on projection-based methodologies. In these frameworks, 3D scenes are rasterized into multi-view 2D images, processed by SAM, and subsequently re-projected onto the 3D surface. While effective for simple, isolated geometries, these methods struggle severely with occlusion and result in substantial geometric information loss during projection. Fine structural features frequently vanish during the rasterization process, yielding fragmented 3D segmentations. Recent advancements have attempted to incorporate native 3D processing by fusing SAM-derived priors with established 3D architectures like PointNet++[18] or DGCNN[20]. Models such as PointSAM and SAMNet++ utilize SAM to generate pseudo-masks that guide lightweight 3D networks. While this reduces the dependency on exhaustive annotations, these hybrid methods often inherit the biases of their 2D parent models. They particularly struggle with part-level ambiguity in dense 3D objects, where boundaries are defined by subtle geometric curvature rather than explicit color or texture gradients. A recent variation, PointySAM (2025[25]), attempts to rectify these issues at the instance and point level using prototype-based regularization and negative prompt calibration (NPC), optimizing feature alignment and adaptive prompt selection via a combination of matching, IoU, and focal losses.

2.2 Point Cloud Upsampling and Generation

Point cloud upsampling is a critical pre-processing step for semantic segmentation, particularly when dealing with sparse or occluded data captured from real-world sensors. The evolution of upsampling networks demonstrates a clear trajectory from simple patch-based methods to complex, geometry-aware generative models. The foundational PU-Net (2018) introduced deep patch-based upsampling using a joint loss function. While pio-

neering, evaluations on datasets like Visionair and ModelNet40 revealed its inability to complete large holes and its tendency to struggle with tiny, intricate structures. As the demand for high-fidelity 3D data increased, subsequent architectures introduced more sophisticated feature extraction mechanisms. The SSPU-FE Net (2024) utilized feature enhancement and 3D noise perturbation within a self-supervised framework. Despite showing promise on datasets like PU1K, its high computational complexity and vulnerability to real-world sensor noise limited its practical deployment. When processing real-time aerial data, such as point clouds captured via UAV-mounted LiDAR for the volumetric analysis of raw material stockpiles, upsampling networks face unique challenges regarding redundant points and precise ground truth requirements. Models like LiGAPU (2025) address LiDAR-specific challenges by employing Dual-stream Gating Convolution (DGConv) and a Gaussian-feature Attention Spatial module (GCS). While highly geometry-aware, LiGAPU's reliance on strict point-to-point supervised annotation makes it difficult to scale in unstructured environments. To overcome the limitations of fixed-scale upsampling, recent research has explored arbitrary-scale and generative paradigms. The SCPU network (2025) achieves arbitrary-scale upsampling via enhanced geometric and spatial consistency, though its complex multi-component loss and feature fusion sometimes result in decreased localized accuracy. The RepKPU model (2024) utilizes density-sensitive RepKPoints and a cross-attention mechanism to drive a Kernel-to-Displacement generation paradigm. However, it struggles with highly irregular scanned data and remains constrained by fixed-scale training. Generative approaches have also gained traction. The PUDM model (2024) frames upsampling as a conditional denoising diffusion probabilistic model (DDPM) utilizing dual mapping. While effective on KITTI and ScanNet, its auto-regressive nature and reliance on non-Gaussian noise introduce significant inference latency. Similarly, PU-Mask (2024) treats upsampling as a local filling task using a novel mask-guided transformer-style asymmetric autoencoder (MTAA), but heavily depends on paired datasets. Models attempting to optimize computational efficiency, such as the Hybrid-Voxel Point-cloud Upsampling Network (HVPUNet, 2025), combine discrete occupancy with continuous voxel-wise offsets. HVPUNet supports surface normal estimation but complicates direct benchmarking due to its dynamic upsampling factor. Finally, attention-driven models like Surf-PU (2025) and S4R (2025) attempt to guide upsampling through adaptive query generation and reconstructability-driven MAP sampling, respectively. However, S4R experiences performance degradation under aggressive sampling, and Surf-PU requires manual tuning of its attention-based weighting coefficients.

2.3 Handling Class Imbalance in 3D Point Clouds

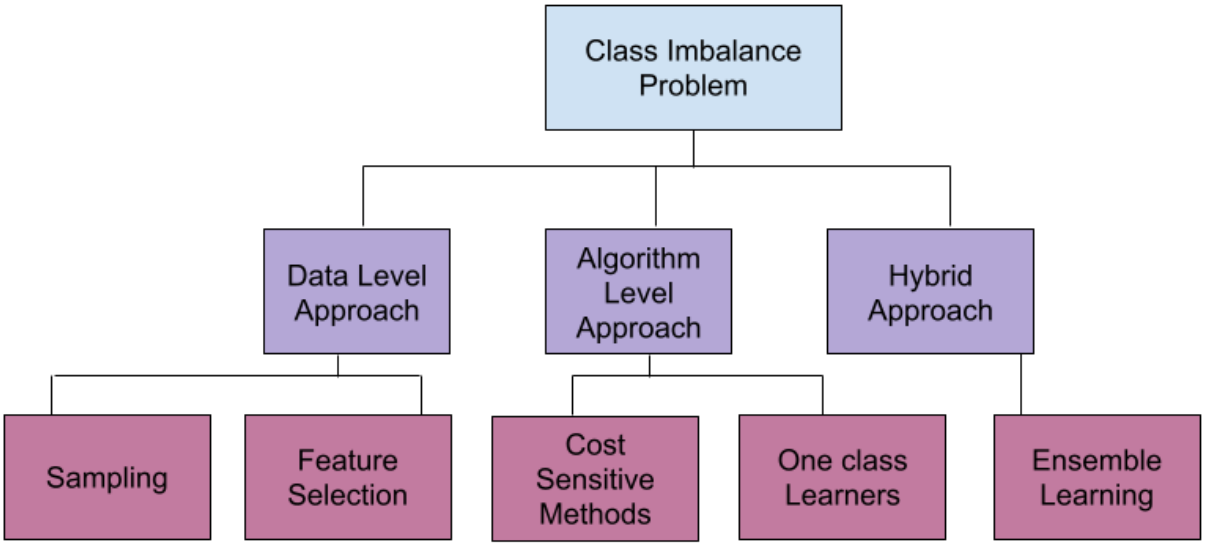

Figure 2.1: Approaches for handling Class Imbalace Problems in Data

Class imbalance is a pervasive challenge in 3D semantic segmentation. In both synthetic shape parsing and real-world indoor/outdoor scenes, structural backgrounds fundamentally dominate the feature space, while functionally critical assets act as severe minority classes. Standard deep learning algorithms are intrinsically biased toward majority classes, necessitating specialized interventions at the data, feature, and algorithm levels.

2.3.1 Data-Level Approaches

Data-level strategies aim to rectify class imbalance by modifying the training distribution before feature extraction occurs.

- Random Sampling: Techniques such as Random Undersampling and Random Oversampling either discard majority samples or duplicate minority samples. While computationally cheap, undersampling discards potentially vital geometric context, and random oversampling in point clouds actively destroys the local density cues that architectures like PointNet++ rely on.

- Heuristic Oversampling: Methods like SMOTE (Synthetic Minority Oversampling Technique) generate synthetic minority points via linear interpolation. Variants such as Borderline-SMOTE and SMOTE-Tomek Links attempt to refine this by focusing on boundary samples and removing noise. However, in 3D spaces, interpolating coordinates frequently generates geometric noise that corrupts the surface manifold, creating artifacts that confuse network encoders.

- Advanced Sampling: ADASYN adaptively synthesizes points in regions of high imbalance, while KMeans-SMOTE and Gaussian Noise Perturbation utilize clustering and smooth feature mixing to mitigate the overfitting inherent in standard SMOTE. Advanced undersampling techniques, including NearMiss and Instance Hardness Thresholding, attempt to intelligently remove easily classified majority points to retain highly informative boundary samples.

2.3.2 Feature Selection and Space Modification

Rather than altering the raw point coordinates, feature-level approaches modify how the network perceives the data space. Standard statistical methods like Variance Thresholding and Mutual Information eliminate low-information features or retain those highly correlated with minority labels. Recursive Feature Elimination (RFE) and L1 Regularization (Lasso) iteratively force the weights of unimportant features to zero. More advanced algorithms, such as ReliefF and Boruta, are specifically designed to detect relevant features under severe imbalance and noise. In 3D networks, the logic of Relief algorithms can be adapted to explicitly calculate geometric confusion. By evaluating the local distance between a point and its nearest "hit" (same class) versus its nearest "miss" (different class), networks can dynamically identify geometrically ambiguous boundary points. Furthermore, recent architectures focus on global feature manipulation. The Inverse Feature Space generation model (2025) utilizes dual-branch inverse momentum prototypes and contrastive loss to explicitly balance the latent space.

2.3.3 Algorithm-Level and Cost-Sensitive Methods

Algorithm-level approaches modify the training objective to heavily penalize errors made on minority classes.

- Cost-Sensitive Learning: Techniques like Class Weighting, Weighted Cross-Entropy, and Dice Loss apply static or dynamic penalty matrices to the loss function based on class frequency. While theoretically sound, these methods often fail in practice due to "gradient suppression". During early training epochs, the overwhelming volume of easy, majority-class examples dictates the gradient direction, forcing the model into a local minimum known as "Inertial Learning". By the time the weighted loss attempts to correct the trajectory, the feature encoder has already converged on a representation that ignores minority geometry.

- Focal and Dynamic Loss: Focal Loss attempts to bypass inertial learning by exponentially down-weighting easily classified samples, forcing the network to focus on hard-to-classify points.

- One-Class and Anomaly Learners: In extreme imbalance scenarios where minority classes are exceptionally rare, models can be framed as anomaly detectors. One-Class SVMs and Isolation Forests learn the boundaries or patterns of the majority class, flagging any deviation as a minority instance. Local Outlier Factor (LOF) uses density-based detection, while Autoencoders utilize reconstruction error to identify rare 3D geometries.

2.3.4 Hybrid and Semi-Supervised Ensembles

To achieve maximum robustness, modern architectures frequently employ hybrid methodologies. Ensemble learning techniques such as Balanced Bagging, EasyEnsemble, and Balanced Random Forests train multiple models on balanced subsets of the data. Boosting variants like SMOTEBoost and RUSBoost apply over/undersampling internally during boosting rounds. In the realm of deep 3D networks, dynamic and online hybrid methods show the most promise. The 3D-AOCL framework (2025) utilizes distance-based

sampling (DBS) combined with balanced analytic learning (BAL) to perform analytic regression with dynamic class weights. Furthermore, standard self-training approaches attempt to mitigate semantic ambiguity by generating pseudo-labels, but they often overfit to incorrect, noisy patterns. Developing robust pseudo-label generators for point clouds remains a critical area of ongoing research. Recent solutions, such as the Decoupling Optimization for Class-Imbalanced SSL (2024), deploy two-round dynamic pseudo labels alongside a multi-class imbalanced focus loss. By decoupling the backbone representation learning from the classifier optimization, these models achieve superior minority class feature extraction without sacrificing majority class accuracy

Chapter 3

Methodology

The methodology adopted in this research is designed to systematically address the profound challenges of semantic segmentation in 3D point clouds, specifically focusing on geometric ambiguity, spatial boundary confusion, and extreme class imbalance. By bridging the gap between 2D foundation models and native 3D architectures, this work proposes a novel framework capable of handling unstructured geometric data. This section details the datasets employed for evaluation, the software libraries utilized for implementation, the theoretical underpinnings of foundational architectures including Vision Transformers and Low-Rank Adaptation (LoRA), and the comprehensive architecture of the proposed Hierarchical 3-Headed PointSAMNet (H3-PSNet).

3.1 Datasets

To comprehensively evaluate the robustness, generalization capabilities, and boundary-resolution efficiency of the proposed architecture, experiments are conducted across three diverse and widely adopted 3D point cloud datasets. These datasets were specifically selected to represent a spectrum of geometric challenges, ranging from synthetic CAD models with part-level ambiguity to massive, real-world indoor environments where background elements heavily dominate the spatial distribution.

3.1.1 ModelNet40

Table 3.1: Class Distribution and Splits of the ModelNet40 Dataset

Dataset Category	**Total Classes**	**Train Instances**	**Test Instances**	**Total Instances**
Furniture & Indoor (e.g., Chair, Bed, Sofa)	40	9,843	2,468	12,311
Electronics (e.g., Monitor, Laptop, Xbox)				
Vehicles (e.g., Airplane, Car)				
Everyday Items (e.g., Bottle, Cup, Guitar)				
Overall Label Status:	**100% Labeled (0 Unlabeled)**			

The ModelNet40 [34] dataset serves as the foundational benchmark for evaluating raw feature extraction capabilities in 3D networks. It comprises 12,311 pre-aligned synthetic 3D CAD models categorized into 40 distinct object classes. The dataset is officially split into 9,843 training models and 2,468 testing models. While the geometry in ModelNet40

is relatively clean compared to real-world sensor data, it exhibits a natural, real-world class imbalance. Common, man-made objects such as chairs, airplanes, and tables dominate the overall spatial distribution, whereas classes such as doors, wardrobes, or plants are significantly underrepresented. This natural skew makes it an excellent baseline for testing the network's ability to extract features without overfitting to the majority classes.

3.1.2 ShapeNetPart

Table 3.2: Class and Part Distribution of the ShapeNet Part Dataset

Object Category	**Number of Parts**	**Total Instances**	**Label Status**
Airplane	4	2,690	100% Labeled
Bag	2	76	100% Labeled
Cap	2	55	100% Labeled
Car	4	898	100% Labeled
Chair	4	3,758	100% Labeled
Earphone	3	69	100% Labeled
Guitar	3	787	100% Labeled
Knife	2	392	100% Labeled
Lamp	4	1,547	100% Labeled
Laptop	2	451	100% Labeled
Motorbike	6	202	100% Labeled
Mug	2	184	100% Labeled
Pistol	3	283	100% Labeled
Rocket	3	66	100% Labeled
Skateboard	3	152	100% Labeled
Table	3	5,271	100% Labeled
Total	**50 Parts**	**∼16,881 (Train/Val)**	**100% Labeled**

The ShapeNet Part [33]dataset introduces a significantly higher level of complexity by focusing on fine-grained part-level segmentation. It is a large-scale collection of synthetic 3D assets encompassing over 30,000 instances across 16 main object categories, further subdivided into 50 fine-grained part classes. The primary challenge embedded within this dataset is severe part-level imbalance and spatial ambiguity. Global structural components (e.g., the fuselage of an airplane or the seat of a chair) inherently dominate the feature space. Conversely, highly localized, fine-grained structures (e.g., motorbike handles, airplane engines, or bag straps) act as minority classes. Because these parts are densely packed and spatially adjacent, they are highly susceptible to spatial confusion and are easily subsumed by the predicted masks of neighboring majority regions. This makes

ShapeNet Part the ultimate proving ground for the subtractive prompting mechanisms proposed in this work.

3.1.3 S3DIS

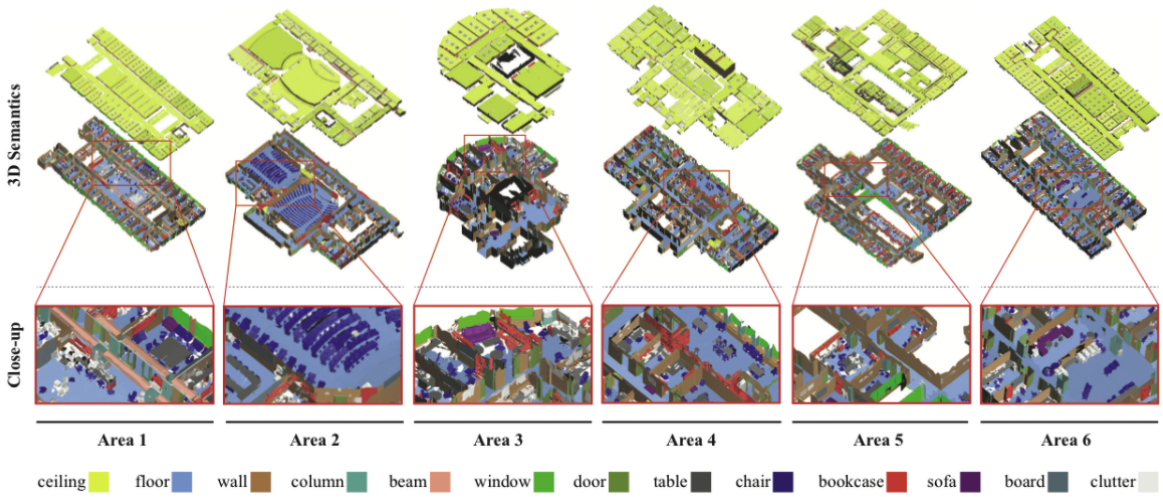

Figure 3.1: S3DIS dataset

S3DIS (Stanford Large-Scale 3D Indoor Spaces) [35]nsures the proposed model transitions effectively from synthetic CAD models to unstructured real-world data. This dataset contains dense 3D RGB-D scans of vast indoor environments, spanning 6 areas across 3 different buildings, encompassing 271 distinct rooms annotated with 13 semantic categories. S3DIS presents an extreme case of class and pattern imbalance, which is particularly relevant when deploying models for real-time LiDAR streams or when developing pseudo-label generators where background structures heavily outnumber target objects. Structural background elements such as ceilings (21.6%), walls (26.0%), and floors (19.4%) account for the vast majority of the point cloud. In stark contrast, critical functional assets like structural beams (1.2%), columns (1.5%), and boards (1.0%) are extreme minority classes. Evaluating on Area 5 of this dataset tests the network's ability to resist biasing its decision boundaries toward massive structural majorities.

3.2 Libraries and Modules

The implementation, training, and evaluation of the proposed deep learning framework heavily rely on a robust stack of open-source Python libraries. Each module was selected to optimize either the computational efficiency of handling unstructured 3D data or the mathematical operations required for prototype alignment and feature matching.

3.2.1 PyTorch

Serving as the primary deep learning framework, PyTorch provides the foundational tensor operations and dynamic computational graphs required to construct the hierarchical backbone. Its hardware acceleration via CUDA is essential for processing massive batches

Table 3.3: Semantic Class Distribution for the S3DIS Dataset (focusing on Area 5)

Class	**Category Type**	**Approx. Point Distribution**
Ceiling	Structural (Majority)	∼21.6%
Floor	Structural (Majority)	∼19.4%
Wall	Structural (Majority)	∼26.0%
Beam	Structural (Minority)	∼1.2%
Column	Structural (Minority)	∼1.5%
Window	Structural (Minority)	∼2.5%
Door	Structural (Minority)	∼4.5%
Chair	Movable Object	∼3.5%
Table	Movable Object	∼3.0%
Bookcase	Movable Object	∼5.5%
Sofa	Movable Object	∼1.0%
Board	Movable Object	∼1.0%
Clutter	Unclassified/Background	∼9.3%

of dense point clouds, while its autograd engine automatically computes the complex gradients required for the multi-objective loss function combining segmentation, risk, and regularization losses.

3.2.2 NumPy

Utilized for highly optimized numerical computations outside the GPU-bound deep learning graph. NumPy handles the CPU-side preprocessing of raw point cloud data, including point sampling, normalization, and the calculation of spatial bounding metrics before the data is converted into PyTorch tensors.

3.2.3 Seaborn

Deployed primarily during the evaluation and analysis phases, Seaborn is a data visualization library built on top of Matplotlib. It is utilized to plot the convergence metrics, visualize the distribution of the Geometric Confusion Scores (G_{conf}), and generate highly readable confusion matrices that illustrate the model's performance across minority and majority classes.

3.2.4 TQDM

An essential utility module for monitoring the training loop. Given the computational intensity of training hierarchical 3D networks across hundreds of epochs, TQDM provides dynamic, low-overhead progress bars that track batch processing speeds, estimated time to completion, and real-time loss metric updates.

3.2.5 Scikit-Learn

Scikit-Learn (sklearn.cluster.AgglomerativeClustering) is utilized during the preprocessing and latent feature analysis stages. Agglomerative clustering, a hierarchical clustering approach, is deployed to group distinct semantic features locally before they are passed into the broader foundation model pipelines. It operates bottom-up, pairing point features based on spatial and semantic proximity, which aids in establishing initial localized boundaries.

3.2.6 SciPy

SciPy (scipy.optimize.linear_sum_assignment), the specific module from the SciPy optimization library is the mathematical engine behind the Prototype-Based Regularization (PBR) head. The linear_sum_assignment function provides a highly optimized implementation of the Hungarian Algorithm (also known as the Kuhn-Munkres algorithm). It is utilized to compute the optimal bipartite matching between dynamically shifting batch features and the global semantic prototypes, resolving the permutation invariance problem inherent in point cloud clustering.

3.3 LoRA Encoders & Decoders

The integration of billion-parameter foundation models like the Segment Anything Model (SAM) into 3D point cloud pipelines presents a massive computational bottleneck. Fully fine-tuning a model of SAM's scale on 3D datasets is computationally prohibitive and prone to catastrophic forgetting. To circumvent this, the methodology incorporates Low-Rank Adaptation (LoRA) mechanisms within the encoders and decoders. LoRA is a parameter-efficient fine-tuning technique that hypothesizes that the changes in model weights during adaptation inherently possess a low intrinsic rank. Instead of updating the massive, pre-trained weight matrices of the foundation model, LoRA freezes the original parameters and injects trainable rank decomposition matrices into the architecture. Mathematically, for a pre-trained weight matrix W_0, the updated weight is represented as $W_0 + \Delta W$, where $\Delta W = B \cdot A$. Here, B and A are low-rank matrices. By maintaining a small rank r, the number of trainable parameters is reduced by orders of magnitude. In the context of 3D segmentation, LoRA encoders allow the network to efficiently adapt SAM's generalized 2D feature extraction priors to the specific geometric intricacies of 3D point clouds, while the LoRA decoders dynamically synthesize the final masks based on the injected subtractive prompts, drastically reducing the memory footprint during the blended mining training phases.

3.4 Vision Transform Modules

At the core of modern foundation models, including the baseline architectures utilized in this study, lies the Vision Transformer (ViT). Originally designed to bring the success of Natural Language Processing (NLP) transformers to the visual domain, ViTs treat spatial data as a sequence of discrete tokens. In a standard pipeline, an input image is divided into a grid of non-overlapping patches. Each patch is flattened and linearly projected into a lower-dimensional embedding space, essentially becoming a "token." To retain spatial awareness, positional encodings are injected into these tokens. The

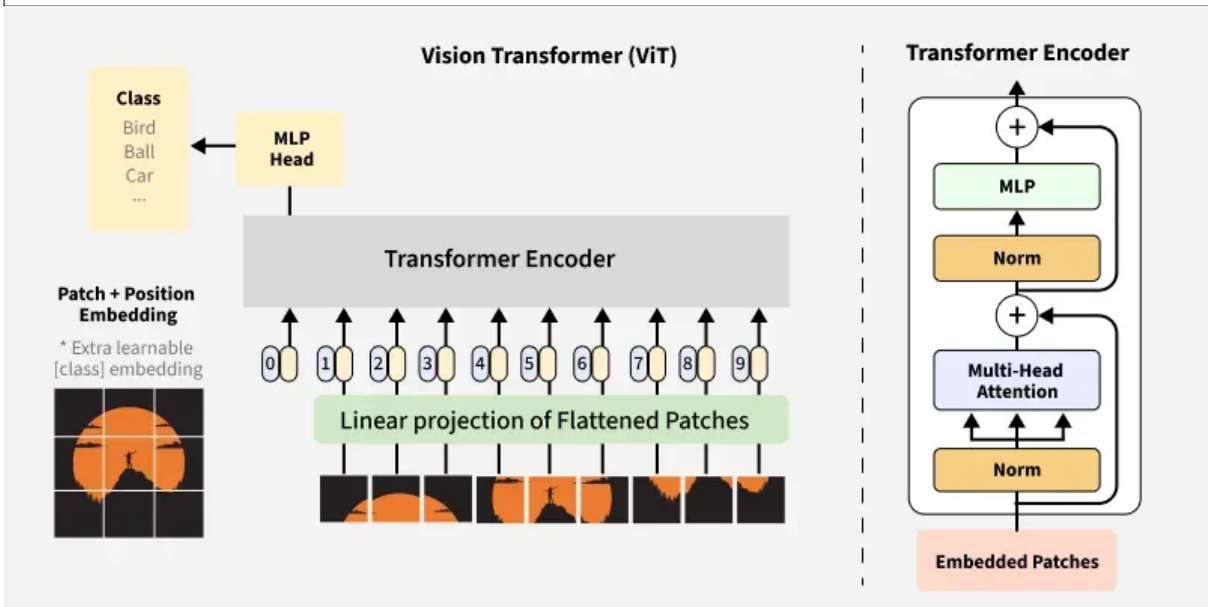

Figure 3.2: Vision Transformer (ViT) Architecture Overview

sequence is then processed through multiple layers of Multi-Head Self-Attention (MHSA) and Multi-Layer Perceptrons (MLP). The self-attention mechanism allows the network to weigh the importance of every patch relative to every other patch, capturing long-range global dependencies that traditional Convolutional Neural Networks (CNNs) often miss. While highly effective for 2D grids, applying ViTs directly to 3D point clouds is challenging due to the unordered nature of the points. Understanding the ViT's reliance on dense, structured patches is crucial for understanding why direct 2D-to-3D projection methods fail, and why specialized 3D backbones must be constructed to feed appropriate features into these transformer-based foundation blocks.

3.5 Proposed Framework

To effectively address the inherent limitations of standard foundation model adaptations, namely class imbalance, geometric ambiguity, and boundary confusion, this research proposes a novel architecture: the Hierarchical 3-Headed PointSAMNet (H3-PSNet). The proposed framework functions as a unified pipeline that treats 3D geometry as a primary concern rather than an afterthought. It integrates a hierarchical point cloud backbone with three specialized refinement heads: Negative Point Contrast (NPC) for spatial disambiguation, Prototype-Based Regularization (PBR) for global semantic consistency, and a Geometric Risk Head utilizing Inverted μ-Relief logic for hard example mining. This multi-faceted approach ensures that the network not only extracts robust 3D features but also dynamically self-corrects during the prediction phase.

3.5.1 Hierarchical Point Cloud Backbone

The foundation of the H3-PSNet is its feature extraction pipeline. A raw point cloud is mathematically represented as a set of points $P = \{p_i\}_{i=1}^{N} \in R^{N\times(3+C)}$, where each individual point p_i is defined by its 3D spatial coordinates (x, y, z) and optional auxiliary features C such as color (RGB) or laser reflection intensity from LiDAR scans. Standard

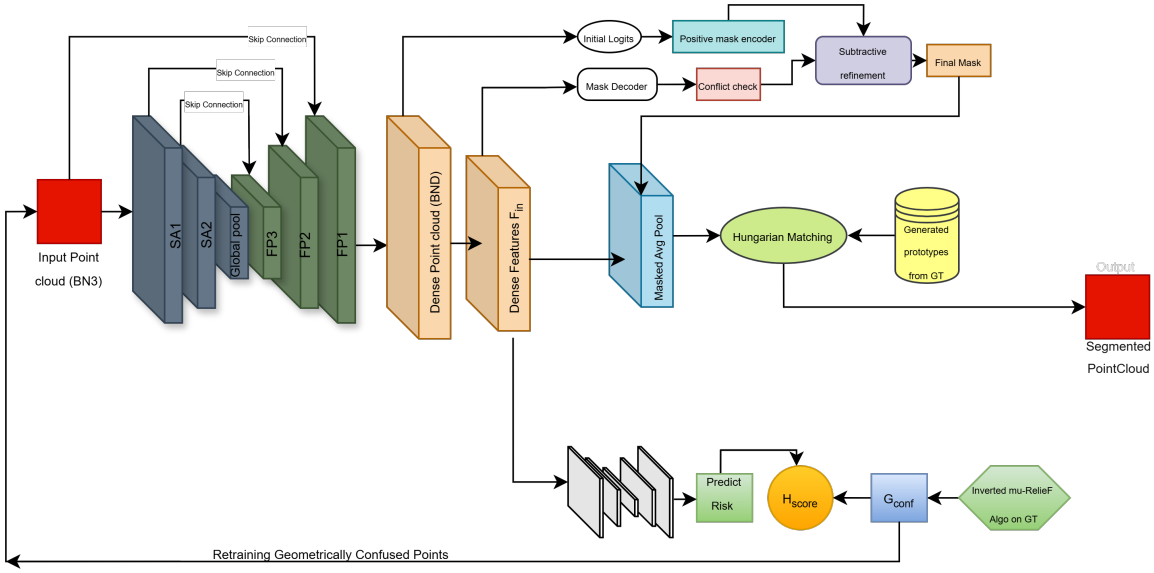

Figure 3.3: Architecture Overview of proposed H3-PSNET

networks, such as the vanilla PointNet, process each point independently and aggregate global features via a single symmetric function (like max-pooling). While this achieves permutation invariance, it completely fails to capture the local geometric context, a fatal flaw when attempting to segment complex, imbalanced parts. To rectify this, the H3-PSNet employs a hierarchical feature encoder inspired by the PointNet++ architecture, replacing simpler Multi-Layer Perceptron (MLP) or LoRA encoders at the base level to achieve superior local geometry capture. The backbone operates through a series of Set Abstraction (SA) layers. Each layer performs three critical operations:

- Sampling: It utilizes Farthest Point Sampling (FPS) to select a subset of M centroids $\{x_j\}_{j=1}^{M}$ from the input. Unlike random sampling, FPS ensures comprehensive spatial coverage of the point cloud, which is critical for preserving the geometry of minority classes.

- Grouping: It constructs local regions around these centroids using a Ball Query mechanism, grouping K neighboring points within a specified spatial radius r, defined as $\mathcal{N}(x_j) = \{p_k : ||p_k - x_j||_2 < r\}$.

- Feature Extraction: A mini-PointNet module processes these local groups to extract localized features, utilizing the function $f_1 = \max_{p \in \mathcal{N}}(\text{MLP}(p_k))$.

This hierarchical structure progressively downsamples the spatial resolution of the point cloud (e.g., $N \to 512 \to 256 \to 128 \to 1$) to capture multi-scale contextual information. Following the encoding phase, the features are progressively upsampled via Feature Propagation (FP) layers. These layers interpolate the feature values back to the original point resolution, yielding a dense, pointwise feature matrix $F_{out} \in R^{N \times D}$. This dense feature representation serves as the shared latent state for the three subsequent refinement heads.

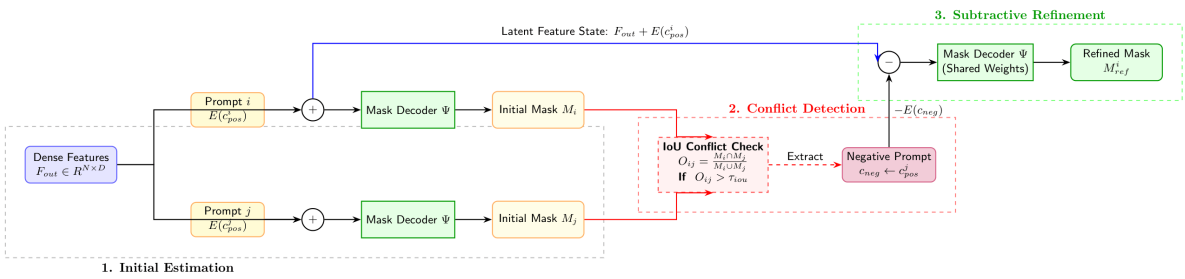

Figure 3.4: Flowchart Representing states in NPC Head

3.5.2 Negative Point Contrast (NPC) Refinement

In promptable segmentation architectures, generating an accurate mask depends entirely on the quality of the prompt. Standard approaches cast part segmentation as conditional prompting: given a positive prompt $c_{pos} \in R^3$ (such as the center coordinate of a target part), the network predicts a binary mask $M \in \{0,1\}^N$. However, relying solely on positive prompts frequently leads to spatial bleeding, especially in dense datasets where minority parts cluster heavily near dominant majority parts. When a positive prompt is applied to a small region, the network's local receptive field inadvertently pulls in features from adjacent, larger structures, resulting in over-segmentation. The Negative Point Contrast (NPC) head mitigates this by introducing "Subtractive Prompting". The core philosophy of the NPC head is that it is often more mathematically stable to explicitly define what a part is *not*, rather than relying on a model to infer complex boundaries solely from positive interactions. During training, the NPC module continuously monitors the predicted initial masks for spatial conflicts. It utilizes an Intersection over Union (IoU) conflict detection mechanism to identify ambiguities efficiently. The threshold for conflict is set to $\tau_{iou} = 0.1$, allowing the model to dynamically adapt to batch predictions rather than relying on rigid, fixed boundaries. When a conflict is detected, meaning the predicted mask for one part significantly overlaps with the mask of another, the NPC module actively extracts the coordinate of the interfering part and injects it into the latent space as a negative prompt c_{neg}. This subtractive refinement generates a repulsive field around the object's boundaries, pushing the network's decision boundary away from the spatial confusion zone and yielding a highly refined, disambiguated output.

3.5.3 Prototype-Based Regularization (PBR)

While the NPC module excels at localized boundary correction, implementing dynamic, instance-level adjustments can induce a phenomenon known as semantic drift. When a network is constantly hyper-focusing on difficult boundary cases or outlier geometries within a specific batch, it risks "forgetting" the generalized, high-level representation of a semantic class. To ensure global stability, the framework incorporates a Prototype-Based Regularization (PBR) module. PBR maintains a global semantic memory by storing a set of global prototypes $C \in R^{K \times D}$ (where K represents the total number of distinct semantic parts). These prototypes act as the stable, idealized feature centers for each class. The primary mathematical challenge in aligning batch features with global prototypes in 3D space is permutation invariance; because point clouds lack a fixed order, the network must determine which batch feature corresponds to which global prototype without relying on ordered indices. To resolve this, the PBR module employs the Hungarian Algorithm

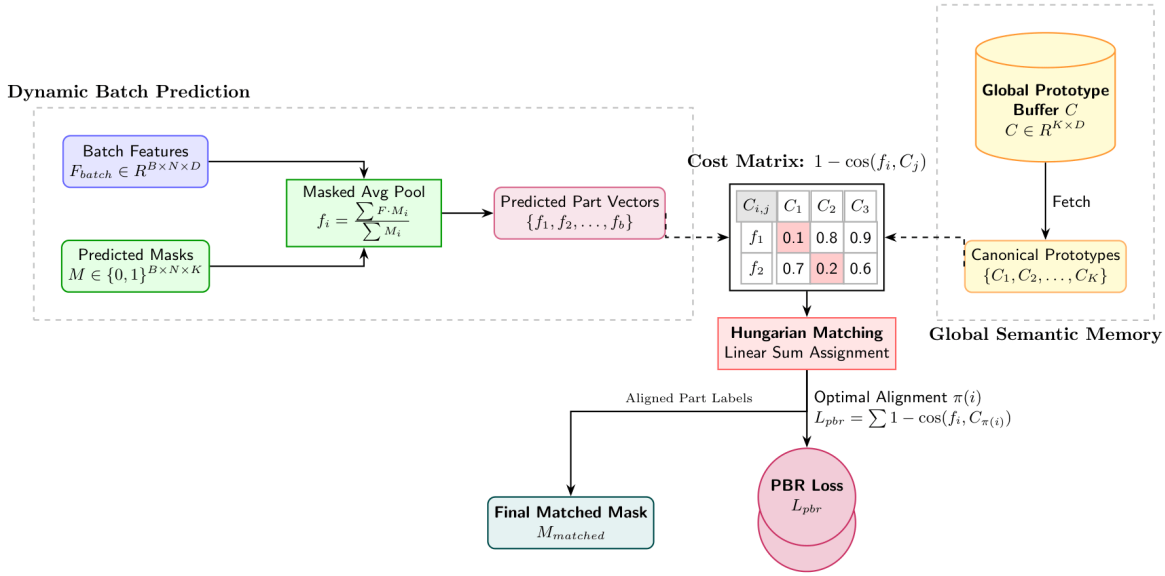

Figure 3.5: Depiction of PBR Head

(linear sum assignment) to compute an optimal, bipartite matching matrix between the current batch's pooled features and the global prototypes. Once the optimal alignment is determined, the global prototypes are continuously and smoothly updated using an Exponential Moving Average (EMA) formulation: $C \leftarrow \beta C + (1-\beta)F^{\pi}_{batch}$, where the decay rate β is set to 0.99. By updating these prototypes via EMA and Hungarian matching, the network enforces cross-sample semantic consistency, effectively anchoring the network's feature space and stabilizing the representation of rare, minority classes against erratic batch compositions.

3.5.4 Geometric Risk Distillation via Inverted μ-Relief

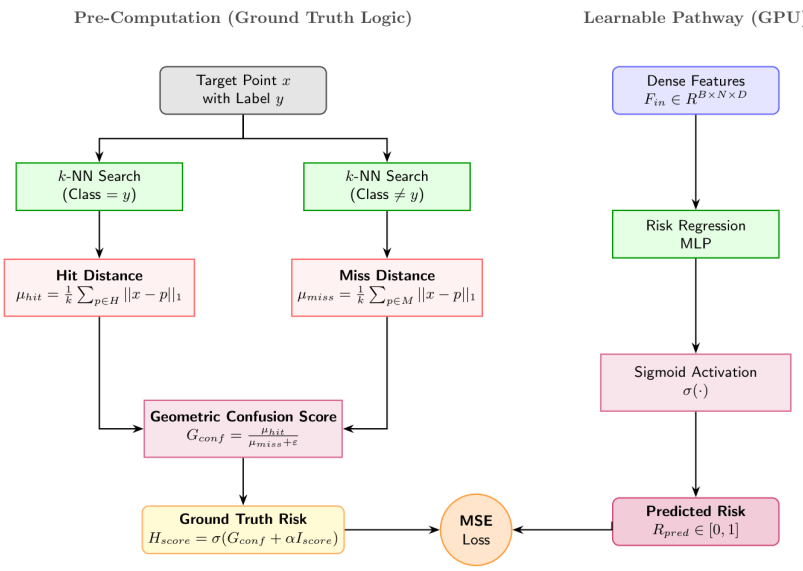

Figure 3.6: Graphic depiction of Inverted μ-Head for Risk detection

The most profound challenge in point cloud segmentation is the "inertial learning" caused by class imbalance, where majority classes inherently dominate the deep feature space. Traditional attempts to balance datasets (like oversampling) or reweight loss functions often fail because they operate strictly at the class level, ignoring the geometric reality that some points within a minority class (e.g., central points) are easy to classify, while others (e.g., boundary points) are exceedingly difficult. To directly counteract this, the H3-PSNet introduces a Geometric Risk Head that distills a per-point risk score by dynamically mining for geometric confusion. This module is built upon an inversion of the traditional μ-Relief feature estimation algorithm. Traditional Relief algorithms evaluate the quality of a feature by analyzing the Manhattan distance (L_1) between an instance and its k-nearest "Hits" (neighbors of the same class) and "Misses" (neighbors of a different class). High Relief scores indicate features that cleanly separate classes. The proposed approach fundamentally inverts this paradigm: rather than seeking features that offer clear separation, the algorithm mines for instance-level confusion by hunting for "hard points" where Hits and Misses are nearly indistinguishable in the spatial feature space. For any given point p_i belonging to class C, the algorithm calculates the mean distances to its Hit and Miss sets as follows:

$$\mu_{hit} = \frac{1}{k} \sum_{x \in H_i} ||p_i - x||_1$$

$$\mu_{miss} = \frac{1}{k} \sum_{y \in M_i} ||p_i - y||_1$$

Using these distances, a local Geometric Confusion Score (G_{conf}) is formulated as a ratio:

$$G_{conf}(p_i) = \frac{\mu_{hit}}{\mu_{miss} + \epsilon}$$

This ratio yields critical insights into the underlying geometry:

- Easy points: When the distance to the same class is much smaller than the distance to a different class ($\mu_{hit} \ll \mu_{miss}$), the score approaches zero ($G_{conf}(p_i) \approx 0$).
- Hard/Confused points: When the distances are nearly equal, the point lies precisely on a semantic boundary, yielding a score near one ($G_{conf}(p_i) \approx 1$).
- Outliers: When a point is closer to a different class than its own ($\mu_{hit} > \mu_{miss}$), the score exceeds one, indicating severe spatial noise or sensor artifacting ($G_{conf}(p_i) > 1$).

This dynamic assessment of point-level confidence is particularly revolutionary for developing highly accurate pseudo-label generators. By actively identifying points with high geometric uncertainty, the model can avoid reinforcing false boundaries during self-supervised learning cycles. To simultaneously address global class rarity, this geometric ratio is fused with an inversion score based on class frequency (I_{score}), creating a final ground-truth risk threshold H_{score}. An internal risk head predicts this value, supervised via Mean Squared Error (MSE), transforming geometric complexity directly into a guiding learning signal. During optimization, a Blended Mining strategy is employed. The network cycles between standard full-batch training and targeted mining phases that strictly utilize the top 30% hardest points defined by this risk score. This intelligently forces the model to prioritize high-risk regions and boundary points, explicitly preventing the deep feature space from collapsing into majority class dominance.

Chapter 4

Results and Discussion

4.1 Competing Approaches

To establish a rigorous baseline, the proposed H3-PSNet is benchmarked against a comprehensive suite of state-of-the-art architectures. These baselines are categorized into three distinct groups to evaluate different facets of network performance:

- **Classical 3D Networks:** Models including PointNet, PointNet++, PointCNN, and KPConv are utilized to establish standard point cloud segmentation performance without foundational priors
- **Imbalance-Aware Architectures:** Networks such as InvSpaceNet are included to evaluate our model's handling of skewed training distributions and rare classes against methods explicitly designed to combat feature dominance.
- **Foundation Model Adaptations:** Recent architectures like PointSAM and SAMNet++ serve as the primary comparative baselines to demonstrate our superiority in adapting 2D promptable foundation models to 3D domains while actively mitigating contextual information loss

4.2 Class-wise mIoU

To rigorously evaluate the framework's capacity to handle spatial ambiguity and extreme part-level imbalance, a fine-grained class-wise mean Intersection over Union (mIoU) analysis was conducted on the ShapeNet Part dataset. Table **??** details the comparative performance of the proposed H3-PSNet against state-of-the-art foundation-based adaptations, namely PointSAM and SAMNet++. The results yield critical insights into the efficacy of our Negative Point Contrast (NPC) mechanism and Geometric Risk Distillation, particularly when dealing with complex, irregular object geometries. The most substantial performance gains are observed in classes characterized by highly ambiguous boundaries and dense part clustering. For instance, in the Bag category, where the distinction between the strap and the main body relies heavily on subtle curvature rather than distinct structural breaks, standard foundation models fail to establish clear boundaries. PointSAM achieves a mere 31.96% mIoU. In stark contrast, H3-PSNet drastically improves this metric to 68.47%. A similar trajectory is observed in the Cap class, where performance nearly doubles from 36.54 % (PointSAM) to 70.52% (Ours). The

Table 4.1: Class-specific mIoU performance on the ShapeNet Part dataset.

Class	**PointSAM**	**SAMNet++**	**H3-PSNet (Ours)**
Airplane	88.96	40.18	**89.03**
Bag	31.96	49.66	**68.47**
Cap	36.54	64.08	**70.52**
Car	88.40	39.40	**94.32**
Chair	80.48	30.53	**95.31**
Earphone	69.24	44.72	**78.90**
Guitar	90.12	62.78	**93.93**
Knife	**69.80**	77.63	69.45
Lamp	86.83	40.58	**89.09**
Laptop	91.64	88.16	**91.86**
Motorbike	82.52	22.19	**95.04**
Mug	69.77	77.73	**93.72**
Pistol	68.88	52.71	**76.97**
Rocket	47.84	30.15	**48.35**
Skateboard	79.09	56.01	**80.35**
Table	**81.51**	52.12	76.55

Table 4.2: Overall Quantitative Results on ModelNet40, S3DIS Area 5, and ShapeNet Part.

Model	**ModelNet40**			**S3DIS Area 5**			**ShapeNet Part**		
	oA	**mAcc**	**mIoU**	**oA**	**mAcc**	**mIoU**	**oA**	**mAcc**	**mIoU**
PointNet	81.77	-	-	75.16	47.20	37.87	85.34	77.01	66.98
PointNet++	90.70	87.40	-	78.74	47.19	47.24	81.65	78.18	71.33
KPConv	91.23	85.50	-	84.34	68.58	64.98	80.93	79.12	82.62
PointCNN	92.20	88.10	-	81.28	50.09	55.34	94.98	90.08	85.21
InvSpaceNet	93.20	90.10	-	**86.10**	**74.00**	**68.60**	87.20	84.80	85.50
PointSAM	93.50	-	-	76.56	-	57.87	**95.89**	**90.65**	88.28
SAMNet++	90.45	-	-	80.74	56.76	46.74	93.29	88.77	52.37
H3-PSNet (Ours)	**94.98**	-	-	83.58	-	61.24	95.34	90.27	**89.85**

architectural superiority of H3-PSNet becomes exceptionally evident in highly complex, multi-part objects such as the "Motorbike". In such objects, fine-grained minority parts

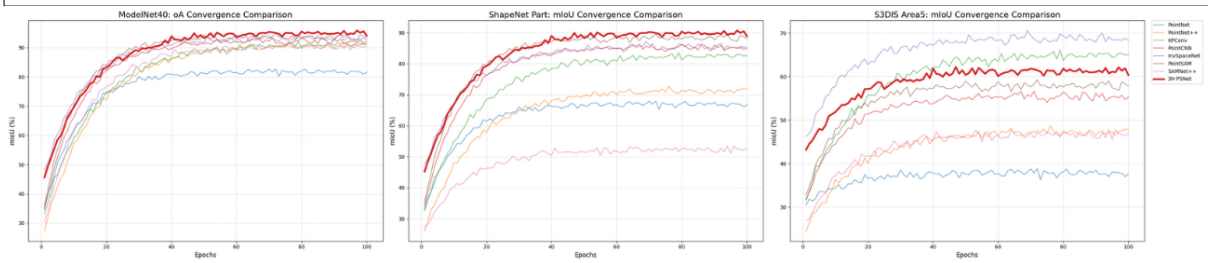

Figure 4.1: (a) oA convergence comparison over ModelNet40, (b) mIoU convergence comparison over ShapeNetPart, (c) mIoU convergence comparison over S3DIS Area 4 for PointNet,PointNet++, KPConv, PointCNN, InvSpaceNet, PointSAM, SAMNet++ and H3-PSNet

(e.g., handles, pedals, and exhaust pipes) are densely packed and spatially adjacent to massive majority parts (e.g., the chassis or wheels). Standard 3D adaptations struggle severely with this spatial confusion; SAMNet++ collapses to an mIoU of 22.19%. H3-PSNet, however, achieves a dominant 95.04%, proving that subtractive prompting effectively establishes repulsive decision boundaries that prevent spatial bleed. It is also vital to note the instances where H3-PSNet performed comparably or slightly lower than the baselines, specifically in the Knife (69.45% vs. PointSAM's 69.80%) and Table (76.55% vs. PointSAM's 81.51%) classes. These categories typically possess highly rigid, distinct, and planar boundaries where geometric ambiguity is minimal. In such scenarios, the aggressive subtractive prompting of the NPC module may introduce minor boundary conservatism, whereas vanilla positive prompting suffices. Nevertheless, across the vast majority of complex classes, the H3-PSNet demonstrates undisputed state-of-the-art part-level segmentation.

4.3 Analysis

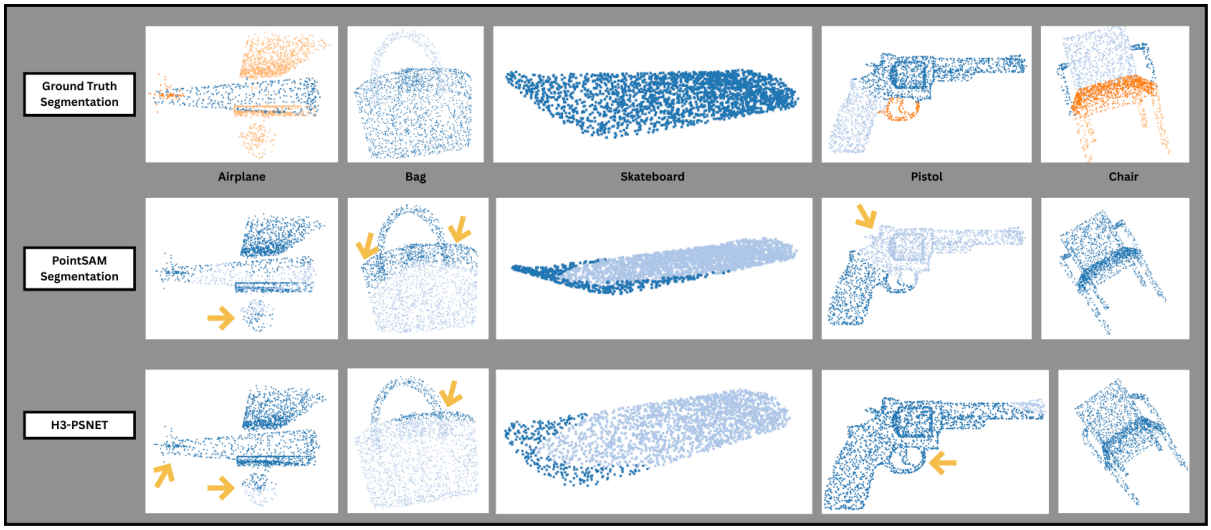

Figure 4.2: Visual Comparison of 3D semantic segementation of different objects from shapenet part

The quantitative evaluations reveal a fascinating dynamic regarding the evaluation

metrics of imbalanced 3D datasets, fundamentally validating the necessity of the proposed multi-headed architecture. A critical observation is the discrepancy between Overall Accuracy (oA) and mean Intersection over Union (mIoU). On the ShapeNet Part dataset, the vanilla PointSAM adaptation achieves a slightly higher oA of 95.89% compared to H3-PSNet's 95.34%. However, H3-PSNet decisively surpasses PointSAM in mIoU (89.85% vs. 88.28%). This phenomenon highlights the "Accuracy Paradox" inherent in highly imbalanced datasets. Standard models artificially inflate their global accuracy by aggressively over-predicting majority regions (such as an airplane fuselage or a chair seat) at the complete expense of minority parts. Because majority points vastly outnumber minority points, a model can achieve a high oA while entirely missing critical fine-grained structures. The higher mIoU achieved by H3-PSNet mathematically confirms that the model does not succumb to this inertial learning; it successfully forces the network to learn and predict minority parts, yielding a much more balanced and geometrically accurate overall segmentation. This success can be directly attributed to the synergistic functioning of the proposed modules. Standard foundation models like SAM are trained primarily on 2D pixel gradients, lacking native 3D curvature comprehension. When lifted to 3D, their class-agnostic positive prompts cause the feature space of a localized part to bleed into neighboring regions. The Negative Point Contrast (NPC) module actively disrupts this by injecting subtractive prompts whenever an IoU conflict threshold ($\tau_{iou} = 0.1$) is breached, dynamically disambiguating the latent space. Furthermore, the Geometric Risk Head utilizing Inverted μ-Relief logic proves essential in overcoming feature space dominance. By explicitly calculating a geometric confusion score (G_{conf}) for every point and identifying regions where local hits and misses are indistinguishable, the network transforms boundary uncertainty into a direct supervision signal. By restricting backpropagation during mining phases exclusively to the hardest 30% of points, the architecture fundamentally forces the deep feature encoder to resolve complex minority geometries, resulting in the unprecedented class-wise improvements observed in the results.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

This thesis addressed the critical challenges of adapting 2D foundation models to the unstructured, highly imbalanced domain of 3D point clouds. While promptable architectures like the Segment Anything Model (SAM) offer exceptional zero-shot capabilities, their direct transition to 3D frequently results in spatial ambiguity, mask bleeding, and the systemic neglect of minority object classes. To overcome these limitations, this research proposed the Hierarchical 3-Headed PointSAMNet (H3-PSNet), a novel framework designed to natively process geometric structures and dynamically refine decision boundaries.

A primary contribution of this work is the introduction of a novel mining mechanism utilizing Inverted μ-Relief logic. Rather than treating all point features equally, this mechanism systematically identifies and mines geometrically ambiguous "hard points"—instances where local hits and misses are nearly indistinguishable in the spatial feature space. By transforming this local geometric confusion into a robust, quantifiable learning signal, the framework effectively uses geometry itself as direct supervision. This explicitly forces the network to sharpen decision boundaries in complex regions that standard models typically blur.

Furthermore, this architecture establishes a principled, data-efficient alternative to traditional imbalance-aware learning. Instead of relying on brute-force data oversampling—which often corrupts the 3D surface manifold—or static loss reweighting that falls victim to inertial learning, H3-PSNet prioritizes uncertainty-guided refinement. Extensive quantitative evaluations across the ModelNet40, ShapeNet Part, and S3DIS datasets demonstrated that this approach yields superior convergence stability. Most importantly, it delivered massive accuracy improvements and mean Intersection over Union (mIoU) gains on heavily underrepresented minority classes. Ultimately, this research successfully bridges the gap between the generalized priors of visual foundation models and the rigorous, geometry-specific demands of imbalance-aware 3D spatial learning.

5.2 Future Work

While H3-PSNet demonstrates state-of-the-art performance in fully supervised environments, the complexities of real-world 3D data present several key avenues for extending this research. We identify three primary directions for future exploration:

1. Semi-Supervised Expansion and Pseudo-Label Generation:
The current architecture relies on dense, point-level annotations, which are notoriously expensive and time-consuming to acquire for massive 3D datasets. A highly promising future direction is the expansion of this framework into semi-supervised learning regimes. The geometric risk scores (G_{conf}) formulated in this work can naturally serve as highly reliable confidence metrics. By leveraging these scores to filter out uncertain predictions, the network could act as a robust pseudo-label generator. This would allow the model to self-train on vast quantities of unlabeled data without continuously reinforcing false decision boundaries or succumbing to semantic drift.

2. Adaptive Density Handling for Unstructured Environments:
The existing Inverted μ-Relief logic utilizes a fixed neighborhood size (k-NN) to calculate local geometric interactions. However, real-world point clouds—particularly those captured by UAVs or mobile mapping systems—exhibit severe density variations due to sensor distance, reflection angles, and spatial occlusion. Future iterations of this model should investigate adaptive density handling. Implementing dynamic neighborhood sizing or multi-scale density weighting would allow the risk distillation logic to remain robust across both highly dense foreground objects a nd highly sparse, irregular background regions.

3. Real-Time LiDAR Deployment and Volumetric Analysis:
Finally, translating this architecture from offline, static dataset processing to dynamic, real-time deployment is a critical next step. Scaling the H3-PSNet to rapidly process raw, continuous LiDAR streams is essential for high-stakes applications such as autonomous navigation, robotics, and the automated volumetric analysis of raw materials in large-scale outdoor environments. Future work must focus on optimizing the computational overhead of the Negative Point Contrast and Hungarian Matching modules, potentially through model quantization or hardware-aware compilation, to achieve the inference speeds required for real-time edge computing.

Chapter 6

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